



Advanced Data Transformation

Module

10 Real-World Deep-Dive Practice Datasets for Power Query Engineers

PIVOT OPERATIONS

UNPIVOT OPERATIONS

PIVOT Rule (Rows → Columns)

Use when you need to build a condensed, aggregate cross-tab summary view.

Memory Trick: "Stretch the table horizontally."

UNPIVOT Rule (Columns → Rows)

Use when wide flat structures prevent proper star-schema filtering and analytics.

Memory Trick: "Stretch the table vertically."

 CASE 01: Sales Analytics

PIVOT

 **Business Use:** Executing clear, cross-tabular Monthly Sales Performance Comparisons.

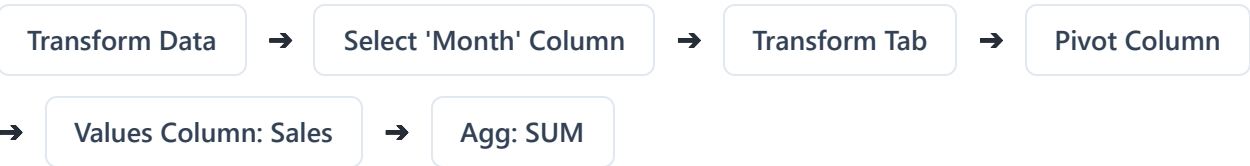
BEFORE TRANSFORMATION

Region	Month	Sales
North	Jan	45,000
North	Feb	52,000
South	Jan	40,000
South	Feb	47,000

AFTER PIVOT COLUMN

Region	Jan	Feb
North	45,000	52,000
South	40,000	47,000

 Power Query Engine Operations:



 CASE 02: Product Sales Pipeline

UNPIVOT

 **Business Use:** Converting human-readable reporting templates into clean, database-normalized structures.

BEFORE TRANSFORMATION

Product	Q1	Q2	Q3	Q4
Laptop	500	650	700	850
Mouse	200	240	310	400

AFTER UNPIVOT COLUMNS

Product	Quarter	Sales
Laptop	Q1	500
Laptop	Q2	650
Mouse	Q1	200
Mouse	Q2	240

 Power Query Engine Operations:

Transform Data



Select Columns 'Q1, Q2, Q3, Q4'



Transform Tab



Unpivot Columns



Rename Attribute/Value Columns

 CASE 03: HR Corporate Payroll

PIVOT

 Business Use: Structuralizing payroll logs into side-by-side comparative monthly department spend.

BEFORE TRANSFORMATION

Department	Month	Salary
HR	Jan	60,000
HR	Feb	62,000
IT	Jan	90,000
IT	Feb	92,000

AFTER PIVOT COLUMN

Department	Jan	Feb
HR	60,000	62,000
IT	90,000	92,000

 Power Query Engine Operations:

Select 'Month'



Transform



Pivot Column



Values: Salary



Advanced: SUM

 CASE 04: Employee Salary Records

UNPIVOT

 **Business Use:** Flattening multiple monthly columns to construct continuous chronological salary trend lines.

BEFORE TRANSFORMATION

EmployeeID	JanSalary	FebSalary	MarSalary
E101	45,000	47,000	48,000
E102	52,000	53,000	55,000

AFTER UNPIVOT COLUMNS

EmployeeID	Month	Salary
E101	JanSalary	45,000
E101	FebSalary	47,000
E102	JanSalary	52,000

 Power Query Engine Operations:

Select 'JanSalary, FebSalary, MarSalary'



Transform



Unpivot Columns

 CASE 05: Student Academics

PIVOT

 **Business Use:** Modeling metrics matrices to conduct immediate semester-on-semester performance evaluation.

BEFORE TRANSFORMATION

RollNo	Semester	CGPA
R101	Sem1	8.2
R101	Sem2	8.5
R102	Sem1	7.8

AFTER PIVOT COLUMN

RollNo	Sem1	Sem2
R101	8.2	8.5
R102	7.8	<i>null</i>

 Power Query Engine Operations:

Select 'Semester'



Transform Tab



Pivot Column



Values: CGPA

 CASE 06: Student Academic Ledger

UNPIVOT

 **Business Use:** Rebuilding normalized data shapes required for cross-filtering dynamic student dashboards.

BEFORE TRANSFORMATION

RollNo	Sem1	Sem2	Sem3
R101	8.2	8.5	8.7
R102	7.9	8.0	8.1

AFTER UNPIVOT COLUMNS

RollNo	Semester	CGPA
R101	Sem1	8.2
R101	Sem2	8.5
R101	Sem3	8.7

 Power Query Engine Operations:

Select 'Sem1, Sem2, Sem3'



Transform



Unpivot Columns

 CASE 07: IPL Sports Analytics

PIVOT

 **Business Use:** Delivering quick-view matrices comparing team scoring performance across specific matches.

BEFORE TRANSFORMATION

Team	Match	Runs
CSK	Match1	180
CSK	Match2	165
RCB	Match1	200

AFTER PIVOT COLUMN

Team	Match1	Match2
CSK	180	165
RCB	200	<i>null</i>

 Power Query Engine Operations:

Select 'Match'



Pivot Column



Values: Runs

 CASE 08: IPL Match Matrix

UNPIVOT

 Business Use: Re-aligning individual match columns into centralized attribute records to isolate score trends.

BEFORE TRANSFORMATION

Team	Match1	Match2	Match3
CSK	180	165	175
RCB	200	150	210

AFTER UNPIVOT COLUMNS

Team	Match	Runs
CSK	Match1	180
CSK	Match2	165
RCB	Match1	200

 Power Query Engine Operations:

Select 'Match1, Match2, Match3'



Transform Tab



Unpivot Columns

 CASE 09: Food Delivery Logistics

PIVOT

 Business Use: Supplying platform executives with localized monthly vendor volume summaries.

BEFORE TRANSFORMATION

Restaurant	Month	Orders
KFC	Jan	350
KFC	Feb	410
Dominos	Jan	500

AFTER PIVOT COLUMN

Restaurant	Jan	Feb
KFC	350	410
Dominos	500	null

 Power Query Engine Operations:

Select 'Month'



Transform



Pivot Column



Values: Orders

 CASE 10: Food Delivery Trend Engine

UNPIVOT

 **Business Use:** Parsing messy commercial spreadsheets into structured granular datasets for rapid timeline analysis.

BEFORE TRANSFORMATION

Restaurant	JanOrders	FebOrders	MarOrders
KFC	350	410	450
Dominos	500	550	580

AFTER UNPIVOT COLUMNS

Restaurant	Month	Orders
KFC	JanOrders	350
KFC	FebOrders	410
Dominos	MarOrders	580

 Power Query Engine Operations:

Select 'JanOrders, FebOrders, MarOrders'



Transform Tab



Unpivot Columns